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***B.Tech. Degree I Semester/Supplementary and II Semester
Supplementary Examination March 2022***

CS/EC/TT 19-200-0203B ENGINEERING GRAPHICS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

(1 × 12 = 12)

- I. Construct a backward reading vernier to show decimeters, centimeters and millimeters and long enough to measure a length of 6 decimeter. RF of the scale is 1/3. Mark 3.69 decimeters and 2.57 decimeters on it.

OR

- II. A disc, diameter 60 mm is rolling on the outside surface of a curved surface of radius 120 mm. Draw and name the path traced by a point 40 mm away from the center of the disc. The point is initially nearer to the center of the base circle.

PART B

(4 × 15 = 60)

- III. The point A of a line AB is in HP and 60 mm in front of VP. The point B is in VP and 40 mm above HP. The distance between the projectors is 70 mm. Draw the projections of the line, find the true length, inclinations and locate its traces. (15)

OR

- IV. Draw the projections of a hexagonal lamina of 30 mm side resting with one of its corners on HP such that the diagonal through that corner is inclined at 30° to VP and the surface of the lamina is inclined at 40° to HP. (15)

- V. (a) A hexagonal pyramid with 30 mm base sides and 70 mm long axis is lying on a slant edge on the ground such that the axis is parallel to the V.P. Draw its projections. (7)
- (b) A cone with a 50 mm base diameter and 55 mm long axis, has a generator in the V.P. and the axis parallel to the H.P. Draw its projections. (8)

OR

- VI. A hexagonal prism of 20 mm base and 60 mm height is resting on one of its corners on the ground with the base making 60° with the ground. The axis is parallel to V.P. A section plane parallel to H.P. and perpendicular to V.P. cuts the object such that it is 15 mm from the base as measured along the axis. Draw its front view and sectional top view. (15)

- VII. A circular hole of diameter 30 mm is drilled through a vertical cylinder of diameter 50 mm and height 65 mm. The axis of the hole is perpendicular to VP and meets the axis of the cylinder at right angles at a height of 30 mm above the base. Draw the development of the lateral surface of the cylinder. (15)

OR

- VIII. A square prism, edge of base 30 mm and height 60 mm, resting on its base on HP, is completely penetrated by another square prism of 20 mm base edge such that the axis of the penetrating prism is perpendicular to and 5 mm in front of the axis of the vertical prism. The rectangular faces of the two prisms are equally inclined to the VP. Draw the projections of the solids showing line of intersection. Assume any suitable length for penetrating prism. (15)

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- IX. Draw the isometric view of the cylinder height 60 mm and base circle diameter 50 mm resting on another cylinder of height 40 mm and diameter 70 mm. The axes of both the cylinders are on the same vertical line. (15)

OR

- X. A square prism of base edge 30 mm and height 60 mm is resting on the ground on a rectangular face with the axis perpendicular to PP and the nearest base 20 mm behind the PP. The station point is 80 mm to the right of the axis of the solid and 50 mm above the GP, 25 mm in front of the PP. Draw the perspective view of the prism. (15)
